INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

HEATER, IMMERSION, LIQUID, FUEL FIRED FOR TANK TRAILERS

This specification is approved for use by the U.S. Army Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a liquid fuel burning immersion type heater designed primarily to warm water in tank trailers in order to prevent freezing of contents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4540

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SPECIFICATIONS

FEDERAL

P-D-680	-	Dry Cleaning and Degreasing Solvent
QQ-A-225/5	-	Aluminum Alloy 2017, Bar, Rod, and Wire; Rolled, Drawn or Cold Finished
QQ-B-654	-	Brazing, Alloys, Silver
QQ-C-390	-	Copper Alloy Castings (Including Cast Bar)
QQ-L-201	-	Lead Sheet
QQ-S-766	-	Steel, Stainless and Heat Resisting, Alloys, Plate, Sheet and Strip
RR-C-271	-	Chains and Attachments, Welded and Weldless
TT-C-490	-	Cleaning Methods for Ferrous Surfaces and Pretreatment for Organic Coatings
TT-E-529	-	Enamel, Alkyd, Semigloss, Low VOC Content
TT-P-664	-	Primer Coating, Alkyd, Corrosion Inhibiting, Lead and Chromate Free, VOC Compliant
WW-P-521	-	Pipe Fittings, Flange Fittings, and Flanges: Steel and Malleable Iron (Threaded and Butt-Welding) Class 150
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-T-60	-	Tape: Packaging, Waterproof

MILITARY

MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank
MIL-P-551	-	Pipes and Elbows, Air Conditioning - Heating
MIL-R-3065	-	Rubber, Fabricated Products
MIL-L-3150	-	Lubricating Oil, Preservative, Medium
MIL-C-10382	-	Corrosion Preventive Petrolatum, Spraying Application: For Food Handling Machinery and Equipment
MIL-C-16173	-	Corrosion Preventive Compound, Solvent Cutback, Cold-Application
MIL-P-19834	-	Plates, Identification, Metal Foil, Adhesive Backed
MIL-L-21260	-	Lubricating Oil, Internal Combustion Engine, Preservative and Break-In
MIL-C-52950	-	Crates, Wood, Open and Covered

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

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MIL-STD-417	-	Classification System and Tests for Solid Elastomeric Materials
MIL-STD-595	-	Colors Used in Government Procurement
MIL-STD-970	-	Standards and Specifications, Order of Preference for the Selection of

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks, are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-13-446	-	Heater, Immersion, Liquid Fuel Fired for Tank Trailer; Assembly, Complete
5-13-447	-	Heater, Immersion, Liquid Fuel Fired for Tank Trailer; Heater Body Assembly
5-13-449	-	Heater, Immersion, Liquid Fuel Fired for Tank Trailer; Fuel Tank Strap Assembly
5-13-453	-	Heater, Immersion, Liquid Fuel Fired for Tank Trailer; Fuel Tank Strap Details
5-13-454	-	Heater, Immersion, Liquid Fuel Fired for Tank Trailer; Collar Assembly
5-13-455	-	Heater, Immersion, Liquid Fuel Fired for Tank Trailer; Hood Assembly
5-13-456	-	Heater, Immersion, Liquid Fuel Fired for Tank Trailer; Fuel Tank Assembly Filler Plug, and Valve Assemblies
5-13-457	-	Heater, Immersion, Liquid Fuel Fired for Tank Trailer; Filler Plug Assembly
5-13-459	-	Heater, Immersion, Liquid Fuel Fired for Tank Trailer; Burner Assembly
5-13-1876	-	Heater, Immersion, Liquid Fuel Fired, for Tank Trailer; Container for Level "A" Shipment

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the

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issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36/A 36M	-	Structural Steel
A 53	-	Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
A 108	-	Steel Bars, Carbon, Cold-Finished, Standard Quality
A 575	-	Steel Bars, Carbon, Merchant Quality, M-Grades
A 663	-	Steel Bars, Carbon, Merchant Quality, Mechanical Properties
A 853	-	Steel Wire, Carbon, for General Use
B 124	-	Copper and Copper-Alloy Forging Rod, Bar, and Shapes
B 633	-	Electrodeposited Coatings of Zinc on Iron and Steel
D 3951	-	Standard Practice for Commercial Packaging
D 3953	-	Strapping, Flat Steel and Seals
F 104	-	Nonmetallic Gasket Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN WELDING SOCIETY (AWS)

Standard Qualification Procedure of the American Welding Society

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P. O. Box 351040, Miami, FL 33135.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Welding Qualifications of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not specified shall be selected by the contractor in accordance with MIL-STD-970. Materials and components specified in accordance with MIL-STD-970 shall not degrade the operational suitability or effectiveness of the heater. The materials used shall be clean and free from any substance, such as dirt, grease, or oils that may be toxic (see 4.4.2.1). The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Steel, carbon.

3.2.1.1 Sheet and strip. Carbon steel sheets and strips shall be of commercial quality.

3.2.1.2 Bar stock. Carbon bar stock steel shall conform to ASTM methods A 108, A 575, or A 663.

3.2.1.3 Wire. Carbon steel wire shall conform to ASTM A 853.

3.2.1.4 Angles. Steel angles shall conform to ASTM A 36/A 36M.

3.2.2 Steel, corrosion-resisting, sheet and strip. Corrosion-resisting steel sheet and strip shall conform to the chemical composition requirements of QQ-S-766.

3.2.3 Brass.

3.2.3.1 Castings. Brass used for the valve body casting shall conform to alloy No. 844 of QQ-C-390.

3.2.3.2 Forgings. Brass used for forgings shall conform to ASTM B 124.

3.2.3.3 Rod. Brass rods shall conform to ASTM B 124.

3.2.4 Aluminum. Aluminum alloy 2017 shall conform to condition T4 of QQ-A-225/5.

3.2.5 Solder. Solder shall conform to QQ-B-654.

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3.2.6 Gaskets.

3.2.6.1 Filler plug gasket, The filler plug gasket shall be fabricated from sheet lead conforming to grade B of QQ-L-201.

3.2.6.2 Vent plug gasket, The vent plug gasket shall be of material conforming to type, class, grade, and suffix letters of SB615AE5 of MIL-R-3065 and MIL-STD-417.

3.2.7 Pipe, outlet. The pipe used in the fabrication of the outlet nipple shall be standard black pipe conforming to ASTM A 53.

3.2.8 Elbow. The malleable iron elbow shall conform to type I of WW-P-521.

3.2.9 Chains.

3.2.9.1 Chain, safety. The safety chain shall be weldless chain, brass, approximately 27 links per foot conforming to type II, class 6 of RR-C-271.

3.2.9.2 Chain, bead pattern. The bead pattern chain shall be a commercial quality free swivel, bead pattern as shown on Drawings 5-13-446 and 5-13-453.

3.2.10 Burner wick. The burner wick shall be soft, highly absorbent roll board in accordance with ASTM F 104.

3.2.11 Hardware. All hardware shall be commercial quality conforming to the material and finish as specified on drawings, parts lists, and documents pertaining thereto.

3.2.12 Pipe stove. The stove pipe shall be a 4 inch diameter conforming to MIL-P-551. Four lengths shall be furnished with each heater.

3.2.13 Primer. Primer shall conform to TT-P-664.

3.2.14 Enamel. Enamel shall conform to class A of TT-E-529. The color shall approximate Olive Drab Color No. 24087 of FED-STD-595.

3.2.15 Identification, instruction and warning plates. The fuel tank identification and instruction plates shall conform to MIL-P-19834. The heater identification, instruction and warning plates shall conform to type III, composition C plate of MIL-P-514 (see 6.4).

3.2.16 Thread sealing compound. The thread sealing compound for the pipe joints shall be a commercial grade anti-seize compound (see 3.7.4 and 4.4.1.1).

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3.3 Design and construction. The design and construction of the heater shall conform in all respects to the drawings listed in Section 2, and all subsidiary drawings and parts lists pertaining thereto, and as specified herein.

3.3.1 Body. The body assembly shall conform to Drawing 5-13-447 and shall not leak when tested as specified in 4.5.1.

3.3.2 Fuel tanks. The fuel tank assembly shall conform to Drawing 5-13-456, and shall not leak when tested as specified in 4.5.2.

3.3.3 Burner plate. The burner plate assembly holes through the top plate, wick, and bottom plate, shall conform to Drawing 5-13-459, and shall be in alignment within ± 0.010 inches.

3.3.4 Filler plug assembly. The filler plug and gasket shall conform to Drawing 5-13-457. The gasket shall be assembled on the filler plug with a press fit to secure the gasket to the plug.

3.3.5 Vent plug and fuel valve. The vent plug and fuel valve shall conform to Drawing 5-13-449 and 5-13-456, and shall have no internal obstruction when tested as specified in 4.5.2.

3.4 Welders. Unless otherwise specified (see 6.2), before assigning any welder to work, the contractor shall provide the Government inspection service with the names of the welders to be employed in the work, together with certification that each of the welders has passed a qualification test as prescribed by either of the following listed codes for the type of welding operation to be performed, and that such qualification is effective as defined by that particular code as follows:

(a) Standard Qualification Procedure of the American Welding Society
(b) Welding Qualification of the American Society of Mechanical Engineers

3.5 Finish.

3.5.1 Zinc plating. Prior to painting, the filler plug assembly without the gasket shall be given a zinc plating finish conforming to type II, class FE/Zn 13 Sc 3 of ASTM B 633.

3.5.2 Copper plating. Copper plating of the burner lighter metal components shall be in accordance with the manufacturer's standard practice.

3.5.3 Priming. Prior to painting the surfaces indicated, the following heater components shall be thoroughly cleaned by any method or type of TT-C-490 and given one complete coat of primer specified in 3.2.13:

a. Heater body - outside surface above the top component.

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b. Top heater body - upper surface and outside and inside surface of rim.

c. Fuel tank - all outside surfaces including filler plug, vent plug, and gasoline valve, while screwed shut, and elbow and nipple screw in place.

d. Hood prior to assembly of instruction plate - all surfaces.

e. Draft gate - all surfaces.

f. Fuel tank strap assembly - all surfaces except male and female threads.

g. Hinge pin and lug pin with cotter pins - all surfaces.

h. Nuts, screwheads, rivet heads assembled in place - all outside surfaces.

Any primer application reaching within 3 inches inside of the burner space or flue will not be considered detrimental.

3.5.4 Enameling. When thoroughly dry the primed surfaces specified in 3.5.3 shall be given a complete coat of enamel specified in 3.2.14. Any enameling application reaching within 3 inches inside the burner space or flue will not be considered detrimental. Touch up of surfaces marred inadvertently will be acceptable.

3.6 Marking for identification and instruction.

3.6.1 Burner and draft gate identification. Burner and draft gate identification shall be indented as specified on Drawings 5-13-455 and 5-13-459. The indentation shall show no sign of penetration, unevenness or distortion on the back side of the burner leg or draft gate.

3.6.2 Fuel tank identification and instructions. Fuel tank identification and instructions as specified on Drawing 5-13-456 shall be printed on a black, nonspecular surface, adhesive backed foil as specified in 3.2.15. Lettering shall be aluminum color.

3.6.3 Heater identification and instructions. Heater identification and instructions shall be as specified on Drawings 5-13-447 and 5-13-454, respectively, and as specified in 3.2.15 (see 6.4).

3.6.4 Warning plate. A warning plate consisting of "WARNING - THIS HEATER MUST NOT BE USED IN INSULATED PLASTIC TANK TRAILERS." shall be printed in 1/4 inch block letters on aluminum photosensitized plate as specified in 3.2.15. The plate shall be located on the top of the heater body assembly opposite the chain-pin and chain-lug assemblies and attached with two No. 6 by 3/8 inch long cadmium plated steel drive screws.

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3.7 Workmanship. The heater shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

3.7.1 Surface defects. The heater shall be clean and free of distortion and misalignment of parts. All parts formed from sheet or strip steel shall be free of warpage, dents, wrinkles, bends not specified, fractures, sharp edges, burrs, and slivers. Castings and molded parts shall be free of sand, fins, pits, blow holes, and sprues. External surfaces shall be free from sharp edges.

3.7.2 Welds. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Spot weld shall be adherent and not burned through. Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed from the finished welds, and rough edges and projections shall be dressed smooth.

3.7.3 Brazing. The surfaces of parts to be brazed shall be free from oxide, scale, paint, grease, and other foreign matter. Brazing shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed.

3.7.4 Threaded fittings. Threads shall not be missing, broken, stripped, or fractured. Threads shall be protected during welding, brazing, cleaning and painting from being distorted, damaged or coated. Thread sealing compound shall be applied to completely cover all threads and any excess after tightening shall be removed (see 3.2.16).

3.7.5 Rivets. All rivets shall be tight and peened with smooth and solid heads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection

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system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall consist of a sample size of the first 100 production units. The first article shall be examined for the defects specified in 4.4.3 and 4.4.4, and leak tested as specified in 4.5.1 and 4.5.2. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the thread sealing compound conforms to the requirement specified in 3.2.16.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

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4.4.2.1 Toxic material. Examination shall be made for existence of any toxic material to determine compliance with 3.2. When nonconformance is noted, correction shall be made to the affected items and process.

4.4.2.2 Zinc plating and priming. Examination shall be made to determine compliance with zinc plating and priming requirements of 3.5.1 and 3.5.3, respectively. When nonconformance is noted correction shall be made to the affected items and process.

4.4.2.3 Leak testing. Each heater body and fuel tank shall be leak tested in accordance with 4.5.1 and 4.5.2. Evidence of failure to comply with specified requirements shall be cause to reject the nonconforming items.

4.4.2.4 Visual examination of operations. Examination shall be made to determine if welding and brazing conforms to 3.7.2 and 3.7.3 and to drawings, application of sealing compound to threads conforms to 3.2.16 and 3.7.4, and fittings are protected from being distorted during brazing and welding in compliance with 3.7.4 and drawings. When nonconformance is noted, corrections shall be made to the items affected and to the operation. Items which cannot be corrected shall be removed from production.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of heaters. The sample unit shall be one heater. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not finished	101	
	Type or color of finish not as specified or blistered, peeled, chipped, or area of no film		201
	Poor touchup, color separation, stained, overspray, or foreign matter imbedded in finish; or otherwise impaired		202
Materials	Component materials not as specified	102	
Construction and workmanship, general (applicable to all components, sub-assemblies and assemblies)	Any component missing	103	
	Fractured, split, punctured, torn, dented, creased, deteriorated, bowed sprung, malformed or miscast	104	
	Component misplaced, or not in proper alignment	105	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship, general (applicable to all components, sub-assemblies and assemblies) (cont'd)	Component loose		203
	Evidence of sharp burrs, slivers,or edges	106	
	Any functioning component that will not operate as intended	107	
	Any functioning component that requires abnormal force to operate		204
	Not connected or joined as specified or poorly accomplished	108	
Welding and brazing (where applicable)	Not type specified, missing, incomplete through burn holes, fractured or adherent	109	
	Slag inclusion, slight undercut, not smooth and uniform, or deposits of scale and flux not removed		205
Soldering	Missing or not adherent	110	
	Incomplete	111	
	Not clean (flux or flux residue not removed) pinholes in solder		206
Bolts, nuts screws, studs and other threaded fastener components	Missing, not type specified, broken, stripped, or fractured	112	
	Components not securely tightened		207
	Excess sealing compound not removed		208
Rivets (when applicable)	Missing, not approved type, loose broken, not peened	113	
Pipe, tubing, fittings and valves	Improperly connected or installed, will not provide a seam where required or stripped	114	
Cleanliness of unit	Not clean		209
Lighter	Missing	115	
Assembly	Any component not properly assembled or secured	116	
	Unit perceptibly out of square or alignment	117	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Assembly (cont'd)	Any assembly or component that does not provide a seal where required	118	
Marking, identification, instructions, warning plate, manual, nameplate, parts lists	Missing, incomplete, or illegible, not specified type or size, not in proper location, not affixed in the prescribed manner, not correct color, not stamped, embossed, printed		
	Vitreous enamels, as specified		210

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of heaters. The sample unit shall be one heater. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size location, sequence, or method of application
Preservation	Improperly applied or missing
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling
	Bulged or distorted container
Content	Number of items required per container is more or less than required

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4.5 Methods of inspection.

4.5.1 Heater body leak test. The heater body shall be submerged in water and filled with compressed air at a pressure of 4.5 ± 0.5 pounds per square inch gage and checked for leaks to determine compliance with 3.3.1.

4.5.2 Fuel tank leak test. The fuel tank with vent plug and fuel valve closed shall be submerged in water and filled with compressed air through the filler hole at a pressure of 4.5 ± 0.5 pounds per square inch gage and checked for leaks to determine compliance with 3.3.2. After testing for leaks the vent plug and valve shall be opened and checked for low of air to determine compliance with 3.3.5.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation.

5.1.1.1 Disassembly. Each heater shall have the fuel tank strap detached from the body of the heater and the cylindrical nuts replaced and finger tightened. The four stovepipe joints shall be detached from the heater and disassembled. The burner shall be removed.

5.1.1.2 Cleaning. All uncoated ferrous metal surfaces of the heater and detached components shall be cleaned with dry-cleaning solvent conforming to P-D-680 and thoroughly dried prior to application of preservative.

5.1.1.3 Application of preservative. The interior of the fuel tank and interior of the body of the heater shall be fogged with preservative conforming to grade 1 of MIL-L-21260 and all excess preservative drained off. All exterior uncoated surfaces of the heater body and fuel tank valve stem shall be coated with preservative conforming to MIL-C-10382. The lighter rod, ring, and staples shall be coated with preservative conforming to grade 1 of MIL-C-16173. The burner shall be lightly coated with preservative conforming to MIL-L-3150 and wrapped in one thickness of greaseproof barrier-material conforming to type II, grade A of MIL-B-121, with the wrap secured by tape conforming to type IV, class 1 of PPP-T-60. The four cleaned stovepipe joints shall be individually coated with medium preservative oil conforming to MIL-L-3150.

5.1.1.4 Unit packing.

5.1.1.4.1 Heater body. The burner shall be suspended and secured within the burner flue with minimum 0.040 inch diameter annealed wire. The hood shall be sealed in place over the burner flue with tape as specified in 5.1.1.3. The preserved lighter shall be placed within the exhaust flue. The draft gate shall be sealed in place with tape as specified in 5.1.1.3.

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5.1.1.4.2 Fuel tank. The outlet pipe and valve assembly of the fuel tank shall be wrapped with greaseproof barrier-material secured in place with tape as specified in 5.1.1.3. The fuel tank strap shall be mounted on the fuel tank and nuts tightened.

5.1.1.4.3 Stovepipe. The rivets used in the stovepipe joints as retaining means shall be placed in a cotton cloth drawstring bag and securely tied to the outer stovepipe through one of the rivet holes. The preserved stovepipe joints shall be nested one within another wrapped with greaseproof barrier-material as specified in 5.1.1.3. The wrap shall completely inclose the stove pipe including the ends and shall be secured in place with the tape as specified in 5.1.1.3.

5.1.1.4.4 Placement. The fuel tank, with tank strap attached, shall be placed on end on top of the lower portion of the body of the heater, centered against the flue stack and the vertical and diagonal braces, and securely held in place by two 3/8 by 0.015 inch flat steel straps conforming to ASTM D 3953. The straps shall pass around the tank and the flue stack and shall be spaced not less than 6 inches apart. Care shall be taken to position the outlet pipe and valve assembly of the tank so that it will not come in contact with the heater or the shipping container. The unit packed stovepipe shall be placed upright beside the vertical brace within the flanged edge of the heater top and securely wired to the vertical brace.

5.1.2 Commercial preservation. Each heater shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or Commercial as specified (see 6.2).

5.2.1 Level A packing. Each heater, complete with components, preserved as specified in 5.1, shall be packed in a nailed wood crate conforming to MIL-C-52950 and the nailed wood container illustration shown on Drawing 5-13-1876; or a wirebound container conforming to PPP-B-585 and the wirebound container illustration shown on Drawing 5-13-1876. Each heater shall be positioned so that the unit packed stovepipe is in one corner of the shipping container. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

5.2.2 Commercial packing. Heaters, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The immersion heater is intended for insertion in a steel or aluminum water tank trailer for the purpose of heating water to prevent freezing and is not intended for use in the insulated water tank trailer. The heater is designed to use standard motor gasoline. It will also operate satisfactorily using any grade of gasoline that may be available. It may be used with kerosene or light diesel oil as fuel, however, its operation will be below par.

6.2 Aquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Source of supply. Photographic sensitized plate is available through the General Services Administration Federal Supply Schedule under FSC Class 6750, Photographic Supplies. Photographic sensitized plate made by Metal Photo of Cincinnati, OH or equal is satisfactory (see 3.2.15).

6.5 Subject term (keyword) listing.

Diesel
Gasoline
Heating water
Kerosene
Water trailer

6.6 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing address for submission of data are listed in FED-STD-313.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodian:

Army - GL

Review activity:

DLA - CS

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-H-52113C(GL)	1992 November 17

3. DOCUMENT TITLE

HEATER, IMMERSION, LIQUID, FUEL FIRED FOR TANK TRAILERS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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